

Primitive-based Model Reduction for Huge 3D Meshes of Industrial Plants

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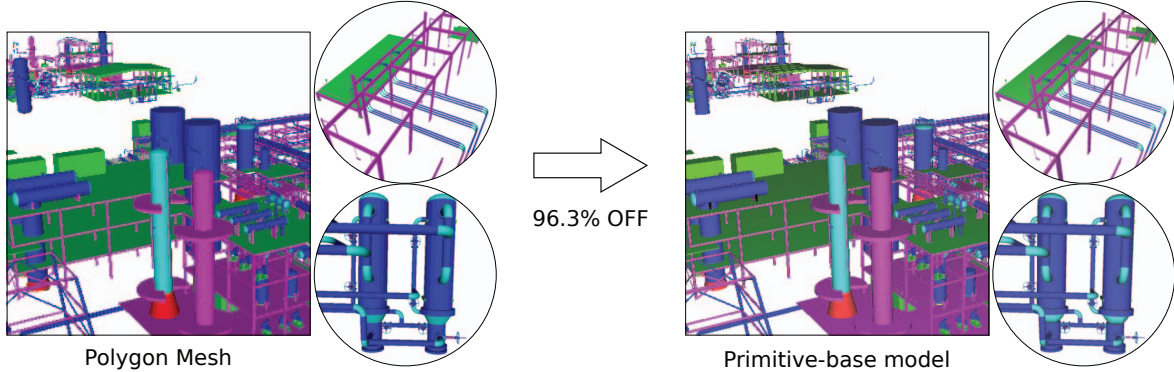


Figure 1: Result of our model reduction using implicit representations of the geometries. The proposed method transforms input 3D polygonal meshes into a set of primitives, each of which is represented by an implicit function. The transformation process consists of the rough part separation, shape classification for each mesh of the part, and implicit function fitting. The visualization in this figure assigns the colors for the types of shapes: green for cuboids, blue for cylinders, light blue for tori and rotational extrusion, yellow for quadratic surfaces, and magenta for those defined by linear extrusion.

ABSTRACT

We address the model reduction for huge 3D meshes of industrial plants, which typically consist of millions of triangles. The proposed method converts an input plant mesh into a set of primitive shapes, each of which is represented by an implicit representation. To this end, we initially partition the input mesh using geometric features, and then, an implicit function is fitted to each part such that the surface deviation is minimized. In this way, the proposed method achieves to reduce the model size to less than 10% of the original one. By sorting the primitives using a bounding volume hierarchy (BVH), our system can also visualize the model-reduced shape format approximately in real-time.

Index Terms: Computing methodologies—Computer graphics—Shape modeling—Mesh geometry models.

1 INTRODUCTION

The recent prevalence of low-cost AR/VR devices has led to their applications for real industries. However, computational resources equipped in AR/VR devices are far less than those equipped in ordinary computers. Therefore, data size reduction is one of the most important problems in facilitating AR/VR applications. While many studies have addressed data reduction for polygonal meshes [2], including recent approaches using deep learning (e.g., the work by Takikawa et al. [3]), the trade-off between the data size and detail preservation remains an ongoing area of research.

Our research targets polygon meshes of industrial plants, which typically consist of millions or billions of triangles. On the contrary, plant meshes are often composed of many simple primitives, such as straight and elbow pipes, flat plats, and conic roofs. Benefiting from such a characteristic of plant meshes, we propose a novel data

Table 1: List of primitives and parameters to define each of them

Primitive Name	Parameters
Sphere	center $\mathbf{c}$, radius r
Cuboid	origin $\mathbf{o}$, three orthogonal directions $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$
Cylinder	center of bottom circle $\mathbf{c}_0$, center of top circle $\mathbf{c}_1$, radii of circles r
Truncated Cone	center of bottom circle $\mathbf{c}_0$, center of top circle $\mathbf{c}_1$, their respective radii r_0 and r_1
Torus	center $\mathbf{c}$, axis of rotation symmetry $\mathbf{n}$, large radius R , small radius r
Elbow	center $\mathbf{c}$, axis of rotation symmetry $\mathbf{n}$, large radius R , small radius r , starting direction of rotation $\mathbf{v}_1$, angle ϕ
Linear Extrusion	direction of extrusion $\mathbf{n}$, length of extrusion h , cross-section polygon $\{\mathbf{p}_1, \dots, \mathbf{p}_N\}$
Rotational Extrusion	axis $\mathbf{n}$, rotation center $\mathbf{c}$, angle ϕ , cross-section polygon $\{\mathbf{p}_1, \dots, \mathbf{p}_N\}$
Quadratic surface	3×3 matrix entries m_{ij} for $i = 1, 2, 3, j = 1, 2, 3$

reduction approach, where the input mesh is first segmented into sub-meshes, which probably correspond to one of the primitives, and then, the most plausible primitive is assigned to each sub-mesh by calculating the primitive parameters in the least-squares manner. The resulting primitives are stored in the leaf nodes of a bounding volume hierarchy (BVH), and the rendering process based on sphere tracing is accelerated.

2 PRIMITIVE-BASED MODEL REDUCTION

Let $\mathcal{M}(\mathcal{V}, \mathcal{E}, \mathcal{F})$ be the mesh model of an industrial plant, where $\mathcal{V}$, $\mathcal{E}$, and $\mathcal{F}$ are the sets of vertices, edges, and faces of the mesh, respectively. Let further $\mathcal{M}_i(\mathcal{V}_i, \mathcal{E}_i, \mathcal{F}_i)$ for $i = 1, \dots, N$ be the i -th sub-mesh of $\mathcal{M}$, where $\mathcal{V}_i$, $\mathcal{E}_i$, and $\mathcal{F}_i$ are vertex, edge, and face sets. Two sub-meshes are disjoint to each other and the union of them becomes $\mathcal{M}$. The proposed method fits the ones in a set of N_{prim} primitives $\mathcal{P} = \{p_1, \dots, p_{N_{\text{prim}}}\}$ for each sub-mesh. The types of primitives and their respective parameters are shown in Table 1.

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Initial mesh partitioning: Initially, the input mesh $\mathcal{M}$ is partitioned into N_{sub} sub-meshes $\{\mathcal{M}_1, \dots, \mathcal{M}_{N_{\text{sub}}}\}$ such that each sub-mesh $\mathcal{M}_j$ has a connected geometry. Then, we assign an optimal primitive $p_i^* \in \mathcal{P}$ for each sub-mesh $\mathcal{M}_i$. The optimal primitive is determined by solving a combinatorial optimization problem. The loss function used in this problem is defined by the average distance between a sub-mesh and the surface geometry of a primitive. Since the vertices $\mathcal{V}_i$ of a sub-mesh $\mathcal{M}_i$ can be sparse, we sample N_{smp} points $\mathbf{v}_{i,k}, k = 1, 2, \dots, N_{\text{smp}}$ on $\mathcal{M}_i$ uniformly. Using these representations, we define the following minimization problem to determine the optimal primitive $p_i^* = \operatorname{argmin}_{p' \in \mathcal{P}} (E_{p'})$ for $\mathcal{M}_i$.

$$E_{p'} = \lambda_1 E_{p'}^1 + \lambda_2 E_{p'}^2, \quad E_{p'}^1 = \min_{\hat{\theta}} \frac{1}{N_{\text{smp}}} \sum_{i=1}^{N_{\text{smp}}} \{h(\mathbf{v}_i; \hat{\theta}^{p'})\}. \quad (1)$$

Here, the function $h(\mathbf{v}_i; \hat{\theta}^{p'})$ means the distance function, it returns the unsigned distance from $\mathbf{v}_i$ to the model p' . Each $E_{p'}^1$ is defined by $\hat{\theta}$ to minimize the average distance, which can be calculated by solving a linear optimization problem. In cases of spheres, the problem can be solved efficiently by matrix decomposition [1]. After determining $\hat{\theta}$, we sample N_{smp} points $\mathbf{v}_{j,k}, k = 1, 2, \dots, N_{\text{smp}}$ on $p'(\cdot; \hat{\theta})$ uniformly to obtain $E_{p'}^2$, the average of nearest distance from model to the mesh. Finally, these weighted sums are the whole loss value $E_{p'}$. We used $\lambda_1 = 0.7$ and $\lambda_2 = 0.3$.

$$E_{p'}^2 = \frac{1}{N_{\text{smp}}} \sum_{j=1}^{N_{\text{smp}}} \{d(\mathbf{v}_j, \mathcal{M}_i)\}. \quad (2)$$

Further decomposition: When fitting extruded shapes still does not obtain a sufficiently small loss value $E_{p'}$, we further decompose the sub-mesh into several segments. To this end, we employ three different segmentation algorithms: region growing (RG) based on dihedral angles and concavity, approximate convex decomposition (ACD) [4], and planar clustering (PC). In PC process, first, planar surfaces are found and they are merged to which is perpendicular to them to obtain extrusion models. Other regions of curved surfaces are chosen to merge or not through the fitting. We empirically find that RG is appropriate for bumpy shapes, ACD is for somewhat more complicated shapes, and PC is for fillets and chamfers.

3 EXPERIMENTS AND RESULTS

Experiment on small-scale plant data: We first examine the proposed method with a small-scale plant mesh, which we denote as “Model-A,” with approximately 1.07 million triangles and 20.8 MB data size. As shown in Fig. 1, Model-A consists of various facilities, such as plumbings, tanks, pillars, and roofs. We evaluate the effectiveness of our method based on resulting data size, surface deviation, and computation time and compare these values with those obtained by the traditional approach using the quadric error metrics (QEMs) [2]. Table 2 shows the model reduction results. The proposed method has successfully reduced the data size to 0.78 MB, which is approximately 3.7% of the original size. Moreover, even though the QEM-based method achieves high data reduction performance and smaller surface deviations, it may inadequately eliminate small components, as shown in Fig. 2. In contrast, our method retains these small parts, while the data size is sufficiently small as that given by the QEM-based method.

Experiment on large-scale plant data: To test our method with large-scale plant data, we synthesize another model, which we refer to as “Model-B,” with 7.75 million triangles and 148 MB data size. For this model, our method has reduced its data size to 3.1 MB, which is around 2.1% of the original data. This result consists of approximately 45 thousand primitives, and the average surface deviation is as small as 0.007% of the diagonal length of its bounding

Table 2: Results of performance analysis on Model-A with about one million triangles. The surface deviations shown here are computed by the average of the distance from the original surface to the surface given after model reduction and those from the reduced model surface to the original surface. The deviation lengths are calculated as a percent against the diagonal length of the model’s bounding box.

Model-A	Original	QEM 20%	QEM 5%	Ours
#polygons	1,066,817	210,627	51,823	—
#primitives	—	—	—	13,126
Size (MB)	20.8	4.2	1.0	0.78
Time (min)	—	0.41	0.47	53.5
Deviation (%)	—	0.004	0.069	0.012

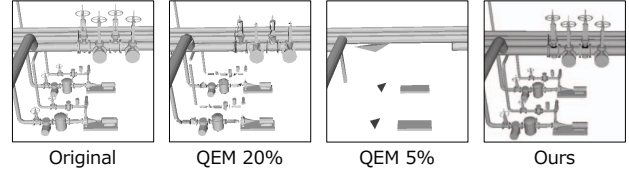


Figure 2: Results of model reduction for a part of Model-A with small structures. These results demonstrate that reduction by QEMs may ignore such small structures.

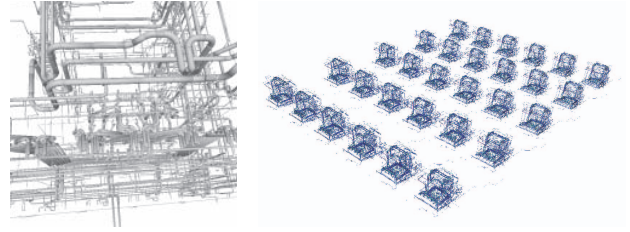


Figure 3: Appearance of Model-B is shown at the left, and a rendering image consisting of 30 copies of that is shown at the right.

box. With this data, we also evaluate the rendering performance of our system. The rendering engine is implemented using C++ and CUDA C and tested on a computer equipped with an NVIDIA GeForce RTX 2060 graphics card. To highlight the efficiency of our rendering system and primitive-based representation, we arrange 30 copies of Model-B in a 5×6 grid and render them all at once into an image with 1024×1024 pixels. When rendering an entire view of the copies, our system achieves 30–50 frames per second (FPS), while it is decreased to 10–20 FPS when the view is zoomed into a specific part of the plant model. In contrast, publicly available mesh viewers, as far as we tested, could not display 30 copies of the original mesh, the data size of which resulted in 4.5 GB, in the same computer as that we tested our method.

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